

UNIT 1 · Introduction to Indian Knowledge Systems

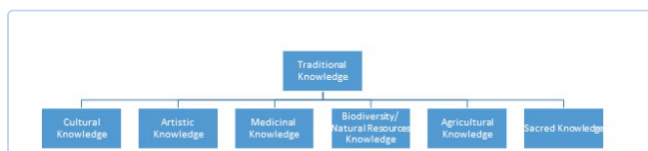
1.1 What is IKS? (Perception & prologue)

- IKS = body of knowledge and practices developed in India over **thousands of years**: philosophy, mathematics, medicine (Ayurveda), arts, architecture, agriculture, understanding of nature. Passed down through generations; promotes **harmony with nature and society**.
- Colonial education marginalised indigenous knowledge; Western epistemology dominated, creating a gap between learners and their roots. **NEP 2020** advocates reviving IKS through curriculum integration at all levels (interdisciplinary learning, indigenous languages, traditional wisdom for sustainability).
- Modern synthesis examples: traditional farming + modern tools; Ayurveda + modern healthcare.
- Languages: Sanskrit carries much of the knowledge; the motto "*Satyam vada, dharmaṃ cara*" (Taittirīya Upaniṣad; SEE MCQ). "*Sā vidyā yā vimuktaye*" = knowledge that liberates.

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1.2 Indian Traditional Knowledge (TK): nature, scope, kinds

- TK / indigenous knowledge = knowledge, know-how, skills, practices **developed, maintained and passed from generation to generation**, often central to a community's cultural/spiritual identity.
- **Scope**: agriculture, science, technology, ecology, medicine (Ayurveda, Yoga, Unani, Siddha, Naturopathy), biodiversity, plus cultural expressions (signs, symbols).
- **How it is expressed**: mostly **oral**: tales, songs, folklore, proverbs, rituals, dance, paintings, carvings; belongs to the community as a group.
- **Kinds of TK (figure)**: Cultural, Artistic, Medicinal, Biodiversity/Natural-resources, Agricultural, Sacred knowledge. (SEE MCQ: "ecological, technical, social".)



Kinds of Traditional Knowledge (your notes)

IMPORTANT TK vs Modern science (MQP

Q1b, CIE-1)

- TK: **holistic, experiential, community-based, oral, sustainable**; knowledge = path to truth (*satya*) and harmony of self–nature–universe.
- Modern/Western: **analytical, reductionist, empirical, documented**; observation + experiment to explain/control nature.

Threats to TK & why protect it

Threats	Need to protect
• Language threat: indigenous languages (e.g. Sanskrit for Ayurveda) decline • External pressures: migration, environment, modern lifestyle, globalisation • Biopiracy: commercial exploitation of natural biochemicals/genetic material (turmeric patent by an MNC) • Colonisation, capitalism/MNCs, consumerism	• Preserve culture & identity • Protect livelihoods • Health benefits (traditional medicine for large populations) • Ecological benefits: e.g. sacred groves conserve biodiversity

1.3 Transition from traditional to Western knowledge **IMPORTANT**

- **Traditional:** Gurukula system; Nalanda, Takshashila; holistic (philosophy + science + arts + ethics); knowledge = *Vidya*, sacred, aimed at **self-realisation and societal harmony**; oral + Sanskrit texts.
- **British rule:** English medium, new curricula, modern institutions; focus shifted to **empirical, utilitarian, job-oriented** learning. Created a divide (an *epistemological shift*; MQP Q2 answer B).
- Silver lining: documentation and global exposure of Indian heritage; today we seek **synthesis** of IKS with modern science.

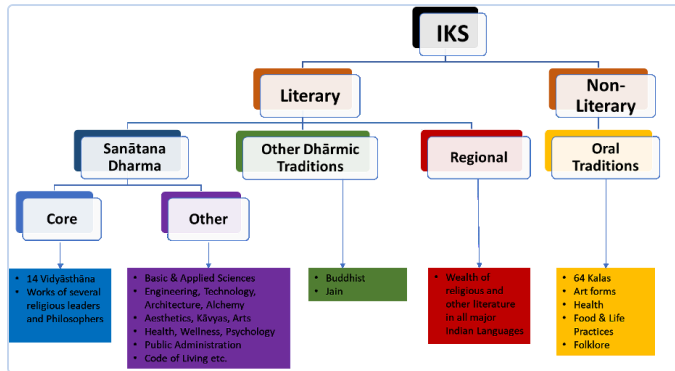
1.4 Historicity of IKS **IMPORTANT** (SEE Feb 2026 1b)

Strand	Key facts
Vedic period (c. 1500–500 BCE)	Origin; four Vedas (cosmology, spirituality, ethics); Upanishads: self (Ātman) and ultimate reality (Brahman).
Epics & classics	Rāmāyaṇa, Mahābhārata (moral guidance); Bhagavad Gītā (in Mahābhārata); Kālidāsa's <i>Śakuntalā</i> ; Patañjali's <i>Yoga Sūtras</i> (holistic well-being).
Philosophy	Six schools: Nyāya, Vaiśeṣika, Sāṃkhya, Yoga, Mīmāṃsā, Vedānta.
Mathematics	Āryabhaṭa, Brahmagupta, Bhāskaračārya: zero, decimal system, trigonometry; Brahmagupta: quadratic equations.
Astronomy	Āryabhaṭa: Earth's rotation, circumference; Varāhamihira: <i>Br̥hat Saṃhitā</i> (astronomy + meteorology).
Medicine	Ayurveda: Charaka Saṃhitā (internal medicine), Suśruta Saṃhitā (surgery).
Learning centres	Takshashila (c. 600 BCE; medicine, law, military science). Nalanda (c. 5th c. CE; library <i>Dharmagaṇja</i> ; students from China, Korea, Central Asia).
Manuscripts & scripts	Palm leaf, birch bark; Brāhmī, Kharoṣṭhī, later Devanāgarī. Architecture texts: Mānasāra, Mayamata, Samarāṅgaṇa-sūtradhāra.

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BOOK p.19 Oral tradition & sūtras ↗

1.5 The IKS Corpus: classification framework **IMPORTANT**



IKS corpus tree (your notes). Memorise this shape.

- **Literary** = **Sanātana Dharma corpus** (Core: Vedas + Vedāṅgas, 14 Vidyāsthānas; Other: basic & applied sciences, engineering, architecture, alchemy, health, psychology, public administration, code of living) + **Other Dhārmik traditions** (Buddhist, Jain) + Regional literature.
- **Non-literary** = oral traditions: 64 Kalās, art forms, health practices, food, folklore.
- Buddhist: Nāgārjuna's *Rasaratnākara* (1st c. CE, alchemy). Jain: mathematics as *Gaṇitānuyoga*, *Tattvārtha Sūtra*.
- Nītiśāstra: **Rāja-nīti** (governance) and **Sāmānya-nīti** (ethics).
- CIE-1 version: Vedic corpus (Śruti) · Post-Vedic (Smṛti: Itihāsa, Purāṇas, Dharmaśāstras) · Traditional sciences · Folk & tribal knowledge.

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[BOOK p.13 Caturdaśa-Vidyāsthāna ↗](#)

1.6 The Vedic Corpus **IMPORTANT**

- **Veda** = "knowledge", from root *vid* (to know). Śruti = "that which was heard": **oral tradition / revelation** (MQP Q1: B). Written form c. 1600 BCE; no definite date as oral transmission was precise.
- Four Vedas = **Chaturveda**: Ṛg, Yajur, Sāma, Atharva.

Veda	Meaning / nature	Facts to remember
R̥gveda	"Praise" · hymns; oldest	1,028 hymns, 10,552 verses, 10 Maṇḍalas ; sūktas: Puruṣa (10.90), Nāsadiya (10.129), Hiranyagarbha (10.121); 5 of 21 śākhās survive (Śākala, Bāṣkala, Āśvalāyana, Śāṅkhāyana, Māṇḍūkāyana). Aitareya Upaniṣad: Mahāvākya <i>Prajñānam Brahma</i> .
Yajurveda	Prose mantras · "book of rituals"	Two branches: Śukla (mantra and brāhmaṇa separate, "bright") and Kṛṣṇa (mixed, "dark") . 101 śākhās once, 6 now (4 Kṛṣṇa: Taittirīya, Maitrāyaṇī, Kaṭha, Kapiṣṭhala; 2 Śukla: Kāṇva, Mādhyandina).
Sāmaveda	Songs / melodies · "book of songs"	Shortest: 1,549 verses ; root of Indian classical music and dance; 3 recensions survive (Kauthuma, Jaiminīya, Rāṇāyanīya); Chāndogya and Kena Upaniṣads.
Atharvaveda	Formulas/spells for diseases and daily life	730 hymns, ~6,000 mantras, 20 books ; Muṇḍaka, Māṇḍūkya, Praśna Upaniṣads; medical/surgical verses (hymn 4.15, fracture).

IMPORTANT **Four divisions of each Veda** (SEE Feb 2026 2b, MQP 2a)

1. **Samhitā**: oldest layer: mantras, hymns, prayers, benedictions.
2. **Brāhmaṇa**: prose commentary on hymns and rituals; how to perform yajña.
3. **Āraṇyaka**: "forest texts": philosophy/symbolism behind the ritual (for hermits).
4. **Upaniṣad**: teacher–student dialogues on the self (Ātman) and Brahman; **Vedānta = end of the Vedas** (CIE-1 MCQ).

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[BOOK p.33 Four divisions ↗](#)

1.7 Vedāṅgas: the six limbs IMPORTANT

Vedāṅga	Subject	Body part
Śikṣā	Phonetics, pronunciation, accent	Nose
Vyākaraṇa	Grammar	Mouth / face
Nirukta	Etymology, meaning of words	Ears
Chandas	Metre / prosody	Feet
Kalpa	Ritual procedure	Hands
Jyotiṣa	Astronomy / timekeeping	Eyes

- **Śikṣā:** Śikṣāvallī chapters: varṇa, svara, mātrā, bala, sāma, santāna.
- **Kalpa sūtras (4):** Śrauta (sacrifices), Gr̥hya (domestic), Dharma (law), **Śulba** (measurement of fire altars; source of geometry).
- **Vyākaraṇa:** Pāṇini c. 400 BCE; Maheśvara Sūtras (14, from Naṭarāja's ḍamaru); Patañjali's *Mahābhāṣya*.
- **Nirukta:** Yāska's *Nighaṇṭu* (c. 600 BCE); "all names come from verbs".
- **Chandas:** Gāyatrī = 3 pādas × 8 = 24 syllables; Anuṣṭup = 4 × 8 = 32 (classical śloka).
- **Jyotiṣa:** Varāhamihira's *Pañcasiddhāntikā* (5 siddhāntas); Pañchāṅga.

IMPORTANT Vedas vs Vedāṅgas: four differences (SEE Feb 2026 1a)

	Vedas	Vedāṅgas
Purpose	Holy texts: origin of world, reason for existence	Ancillary studies to preserve, interpret, perform the Vedas
Structure	4 collections	6 limbs
Content	Hymns, incantations, rituals	Metre, grammar, linguistic analysis, astronomy
Origin	Among the oldest scriptures, c. 1500–1200 BCE	Last treatises of Vedic literature

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[BOOK p.39 Śikṣā ↗](#)

[BOOK p.43 Kalpa ↗](#)

[BOOK p.45 Jyotiṣa ↗](#)

1.8 Indian philosophical systems (Darśanas) IMPORTANT

Origins in Ṛgveda hymns (Nāsadīya, Puruṣa, Vāk) → flowering in Upaniṣads. **Āstika** = accepts Vedas as valid (6 schools). **Nāstika** = denies Vedas (Cārvāka, Bauddha, Jaina).

School	Founder / text	Core idea
Nyāya	Akṣapāda Gautama · Nyāya Sūtras	Logic & epistemology ; 4 pramāṇas (perception, inference, comparison, testimony).
Vaiśeṣika	Kaṇāda · Vaiśeṣika Sūtra (6th c. BCE)	Atomism, pluralism ; physical reality made of atoms; originally 2 pramāṇas (perception, inference).
Sāṃkhya	Kapila · Sāṃkhya Kārikā	Oldest orthodox; dualism of Puruṣa (consciousness) and Prakṛti (matter) ; 3 guṇas (sattva, rajas, tamas); 25 elements.
Yoga	Patañjali · Yoga Sūtras	Sāṃkhya + God; 8 limbs (aṣṭāṅga) ; goal kaivalya.
Pūrva Mīmāṃsā	Jaimini	Interpret Vedas; ritual action (karma-kāṇḍa); salvation through Vedic action.
Vedānta (Uttara Mīmāṃsā)	Bādarāyaṇa · Brahma Sūtras	Upaniṣadic knowledge (jñāna-kāṇḍa). Advaita (Ātman = Brahman), Viśiṣṭādvaita (attributes), Dvaita (3 realities).
Cārvāka (Lokāyata)	Bṛhaspati	Only pratyakṣa valid; "extreme nāstika"; material.
Bauddha	Gautama Buddha	Non-theistic; Four Noble Truths, Noble Eightfold Path to end <i>dukkha</i> .
Jaina	24 Tīrthaṅkaras; Mahāvīra (6th c. BCE)	Anekāntavāda (many viewpoints), ahiṃsā ; only Kevalīs know full truth.

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[BOOK p.65 Mīmāṃsā/Vedānta ↗](#)
[BOOK p.70 Jaina ↗](#)
[BOOK p.72 Bauddha ↗](#)
[BOOK p.75 Cārvāka ↗](#)